

APOLLO SOLAR VENTURES INC.

NUTECH Hardware & Software Solutions

NUTECH School System

Application Documentation

[*portal.html*](#) · [*gate.html*](#) · [*cafeteria-pos.html*](#)

Prepared for

Bernard P. Jocson — Founder & President · Developer: Carlos

Audience: Developers & operators (deployment, configuration, integration)

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Contents

1. Overview

The system has **three browser-based front-ends**, each a self-contained HTML file (HTML + CSS + JavaScript in one file, no build step). They are capture surfaces that all talk to **one n8n workflow** over a single webhook, which reads and writes the shared Airtable base. The card UID is the primary key across all three.

File	Role	Runs on
<code>cafeteria-pos.html</code>	Staffed cashier checkout	POS PC + RFID reader
<code>gate.html</code>	Entrance two-factor attendance	Gate PC + reader + camera
<code>portal.html</code>	Role-based PWA (admin/parent/student/staff)	Phone or PC

Common conventions

- **Single webhook:**
- **Demo vs Live:** each app has a CONFIG flag / toggle. Demo runs fully offline on sample data; Live posts to n8n. It is a chosen mode, not an automatic fallback.
- **Protocol:** each app POSTs JSON ; n8n routes by the value.
- **Browser:** Chrome or Edge, served over or HTTPS (required for Web Serial + camera).

2. cafeteria-pos.html

Purpose

The staffed canteen terminal. The cashier taps a student's card, sees the wallet balance, rings up menu items, takes payment from the wallet, and prints a receipt. Also handles wallet top-ups, linking new cards (enroll), and reassigning a card to a new owner.

Screens / tools

- **Checkout:** tap card → balance + menu → build cart → pay → receipt.
- **Top-up:** add money to a wallet.
- **Enroll:** link a blank card to an account.
- **Reassign:** move a card to a new owner (Admin PIN gated).
- **Settings:** webhook URL, RFID reader (HID / Arduino), demo flag.

Configuration (top of file)

```
DEMO_MODE: false           // POS ships live
N8N_BASE: ".../webhook/cafeteria"
RFID_SOURCE: "hid"         // or "arduino"
ARDUINO_BAUD: 115200
REQUIRE_FP: false         // fingerprint removed
ADMIN_PIN: "1234"          // reassign tool
```

Backend actions used

Action	Sends / does
verify	card tap → returns identity + balance
purchase	charge wallet for the cart; writes sale + line items + stock
topup	add money to wallet
enroll	link a blank card to an account
menu	load menu items
account-info	look up a wallet
reassign	move card to a new owner

Hardware

- **RFID:** Arduino + PN532 over USB (Web Serial). **Confirmed working.**
- **Fingerprint:** removed (DigitalPersona offline) — card-only checkout.
- **Printer (optional):** XP-58IIH via print-bridge on port 9002.

3. gate.html

Purpose

The entrance station. A person taps their RFID card; the app identifies them, then performs a 1:1 face match against the face stored on their ACCOUNT (browser camera + face-api.js). On a match it writes the attendance record — Time In or Out, marked Present or Tardy by the cutoff. It denies entry on a blocked/lost card or a face mismatch.

Flow

1. **Tap** → returns identity, status, the face on file, and today's punches.
2. **Face match** → browser compares the live camera face to the stored descriptor.
3. **Punch** → writes the DTR row + an ATTENDANCE row.

Configuration

```
DEMO_MODE: true
N8N_BASE: ".../webhook/cafeteria"
CUTOFF: "08:00"           // tap after = Tardy
DEVICE: "GATE-01"
USE_CAMERA: false         // real face capture vs simulated
FORCE_FAIL: false         // demo the deny path
RFID_SRC: "hid"           // or "arduino"
ARDUINO_BAUD: 115200
MATCH_THRESHOLD: 0.5      // lower = stricter face match
```

Settings panel (⚙️)

- Demo mode toggle; Use real camera; Force face mismatch (deny demo).
- Tardy cutoff time; device name; webhook URL; Test connection.
- RFID source: USB HID or Arduino (with Connect button + auto-reconnect).

Backend actions used

Action	Does
gate-verify	identity + face on file + today's punches
gate-punch	writes DTR (Time In/Out, Present/Tardy) + ATTENDANCE

Hardware

- **RFID:** USB HID or Arduino + PN532 (Web Serial, auto-reconnect).
- **Camera:** getUserMedia + face-api.js. Needs localhost/HTTPS, Chrome/Edge.
- **Note:** test fixtures have no face stored — keep OFF until faces are enrolled.

4. portal.html

Purpose

The role-based progressive web app (installable, works offline in demo). One file serves four roles, each seeing only their own screens. It is the management and self-service surface — not a tapping station, except the admin Register screen which reads a card.

Roles & screens

Role	Screens
Admin	Dashboard, Register, Cafeteria, Credit Requests, Attendance, People, Reports, Settings
Parent	Wallet view, Credit request (with photo proof)
Student	Wallet + attendance, plus Settings
Personnel	Salary-deduction view, plus Settings

Configuration

```
DEMO_MODE: true           // toggle in Settings
N8N_BASE: ".../webhook/cafeteria"
ADMIN_PIN: "1234"
RFID_SOURCE: "hid"        // or "arduino"
ARDUINO_BAUD: 115200
```

Settings page

- Connect-to-live switch (demo ↔ live); webhook URL; Test connection.
- RFID reader selector (USB HID / Arduino) with Connect + auto-reconnect, POS-style.

Register screen — card capture

In live mode the reader is **auto-armed** on the Register screen: just tap a card and the UID drops into the field (HID or Arduino). In demo mode the Tap button generates a sample UID.

Backend actions used

Action	Feeds
register-person	creates ACCOUNT + role row together
credit-request	parent submits a top-up request with proof
credit-requests	admin lists pending requests
credit-approve	admin approves → wallet credited
admin-overview	dashboard totals
client-data	a person's own history
salary	personnel monthly cafeteria spend
menu / account-info / topup / purchase / order	cafeteria + wallet helpers

5. Running & Deploying

Serve locally (for hardware)

Web Serial (Arduino) and the camera require **localhost** or **HTTPS**. From the folder:

```
python -m http.server 8080
# then open http://localhost:8080/cafeteria-pos.html
```

Going live (per app)

4. **cafeteria-pos.html** — already live; just confirm the webhook in Settings.
5. **gate.html** —  → uncheck Demo mode → Test connection.
6. **portal.html** — Settings → Connect to live system ON → Test connection → Save.

Prerequisite: the n8n workflow must be Active, or Test connection fails with “Cannot reach n8n”.

One reader per page

A USB RFID reader is owned by a single browser page at a time. On one laptop, only one app can hold the Arduino — close the others' tabs. In production each station has its own reader.

6. Quick Reference

File	Ships as	RFID	Camera	Primary actions
cafeteria-pos.html	Live	HID / Arduino	—	verify, purchase, topup
gate.html	Demo	HID / Arduino	Yes	gate-verify, gate-punch
portal.html	Demo	HID / Arduino (Register)	—	register-person, credit-*, dashboards